

OPERATOR'S GUIDE

for

4-Channel Cervical-Thoracic-Lumbar (CTL) Array

for use with

GE Signa® Eclipse 3.0T



GE Catalog Part Number: M1385AB
GE Manual Part Number: 2380634-3

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Manufactured by USA Instruments, Inc.



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SECTION 1 – INTRODUCTION

This manual describes the safety precautions, features, use and care of the 4-Channel CTL Array compatible with the GE Signa Eclipse 3.0T system. Please review this manual thoroughly before using the device.

If you have any questions or comments on this manual, or need any assistance with the use of the product, please contact your GE Medical Systems representative at:

1-800-GE CARES

1-1 Operator's Guide Information

The operator's guide provides information on how to use the 4-Channel CTL Array. The user should read this guide before using the coil for imaging. The user is advised to refer to the MR scanner manual in conjunction with this guide.

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1-2 Use of Symbols

Operators should understand the following symbols used for caution and warning explanations:



A hazard exists which will or could cause a minor injury.



A hazard exists which could cause serious injury.



Attention! Consult accompanying documents.



Type BF equipment.



Do not cross or loop coil cables.

1-3 Coil Classification



Class II equipment.



Type BF applied.

Ordinary equipment.

Suitable for continuous operation.

Medical Device Directive

These products conform with the requirements of council directive 93/42/EEC concerning medical devices, when they bear the following CE Mark of Conformity:



CAUTION:

**Federal law restricts this device to sale, distribution,
and use by or on the order of a physician.**

1-4 Quality Assurance

The procedures described in the Coil Imaging Performance Verification section of this manual should be performed upon receipt of the coil to establish a baseline of coil performance. The procedure should be repeated at regular intervals.

1-5 Safety

This manual contains detailed information on the set-up, positioning and use of your 4-Channel CTL Array. The instructions should be read carefully and thoroughly before attempting to scan patients with the coil.

1-5-1 Coil Operating Safety

Patient safety and comfort should be the primary concern during the scanning procedure. Always follow proper safety, operating and maintenance procedures to ensure that the patient is not exposed to electrical or mechanical hazards that may potentially cause injury.



CAUTIONS:

The following general warning statements apply to scanning with a magnetic resonance system. For further details, review the warnings in your MR system operator's manual.

Cables should not be looped or crossed as arcing and patient burns could result.

Route cables through the center of the magnet bore. Place cables under the cushion whenever possible. Routing near the sides of the bore increases the likelihood of cable heating from induced currents.

Do not allow the surface coil cable to touch the patient; patient burns can result. Use a thermal resistant material or pad to keep the cable from touching the patient.

Keep the length of the cable in the bore to minimum. Avoid bending the cable 180 degrees. And route the cables out of the bore in the most direct way – without looping or coiling.

Visually inspect the cable insulator jackets, strain reliefs and connector boxes before each use. If the insulation is broken, or if the cable is frayed, immediately discontinue use of the device.

Remove unplugged surface coils or unused accessory devices from the magnet bore; a patient burn can result.

Remove blue wire-type ECG gating cables from the magnet bore before scanning with surface coils.

Do not use blue wire-type ECG leads when you are doing gated surface coil scans. When they are used in the presence of a surface coil cable, the potential for heating increases. Use peripheral gating or high impedance ECG leads rather than blue wire-type ECG leads when you are doing gated surface coil scans.

When using any coil, make sure the patient's body does not touch the magnet bore. Use thermal resistant material or padding between the patient and the magnet to prevent burns that could be caused by patient contact with the interior of the magnet bore.

Use only GE or GE-authorized accessory coils and cables.

Use only accessories in good condition. If you suspect that an accessory is not in good condition, discontinue its use and contact your local GE service engineer.

Do not use surface coils with exposed metal conductors or damaged insulation. Skin contact with metal conductors can cause burns.

To help prevent a patient burn from closed loops formed by clasped hands, by hands touching the body, or from thighs or knees contacting over a small area, insert non-conducting pads at least 0.25 inches thick between the touching parts.

1-5-2 Patient Safety



CAUTIONS:

The following general warning statements apply to scanning with a magnetic resonance system. For further details, review the warnings in your MR system operator's manual.

If a patient complains of burning, tingling, stinging, or other similar "burn"-type sensations, promptly stop the scan procedure, examine the patient, and contact the responsible physician before continuing the procedure. Pay special attention to very young, sedated, or other compromised patients who may not be able to communicate effectively.

Patients with ferromagnetic metal should not be scanned because the magnetic field may interact with implanted surgical clips or other ferromagnetic materials.

Persons with cardiac pacemakers or other implanted electronic devices should not enter the magnetic field zone delineated by the MR system manufacturer.

There is a risk to scanning feverish or decompensated cardiac patients.

Facial makeup should be removed before scanning because it may contain metal flakes, which can cause skin and eye irritation. Permanent eyeliner tattoos may cause eye irritation due to ferromagnetic particles.

Patients who work in environments in which there is a risk of embedded metallic fragments in or near the eye should be carefully screened before undergoing an MR exam.

WARNING!

Prevent all cables from forming loops and coming into contact with the patient. Contact may result in burns to the patient.

Ensure that the patient is comfortably positioned.

Monitor the patient periodically. Stop the scan immediately if the patient reports any heating, burning or tingling sensations.

1-5-3 Emergency Procedures

In the unlikely event that a coil creates smoke, sparks or makes an unusually loud noise, or if the patient requires emergency assistance:

- Stop the scan if one is in progress.
- Remove the patient from the scan room if medical treatment is needed.

1-5-4 Indications

The 4-Channel CTL Array is a receive-only RF coil used for obtaining diagnostic images of the cervical, thoracic and lumbar spine regions in Magnetic Resonance Imaging systems. The indications for use are the same as for standard MR imaging. The 4-Channel CTL Array is designed for use with the 3.0T MRI scanner manufactured by GE Medical Systems.

1-5-5 Equipment Safety

All personnel using the coil should be instructed in the proper connection and handling of the coil.

Personnel should observe all warnings and cautions that appear in this guide.

WARNING!

Auxiliary equipment (such as gating equipment, vital signs monitoring systems and RF coils) that have not been specifically tested and approved for use in the MR environment may result in burns or other injuries to the patient. This equipment may also interfere with the proper operation of the coil.

WARNING!

The coil is not suitable for use in the presence of flammable anesthetics in combination with air, oxygen or nitrous oxide.

CAUTION



Prevent cables from forming loops. Looping will degrade scan performance of the coil through RF coupling.

WARNING!

This equipment is not protected against ingress of water.

Note: This equipment is rated for continuous operation.

Note: For the disposal of this coil at the end of its useful life, please contact GE Service Personnel.

1-5-6 Electrical and Mechanical Safety

WARNING!

The coil consists of both electrical and mechanical components. Service personnel must have specialized training to ensure the safe operating condition of the coil. Therefore, only properly trained and qualified personnel should be authorized to service the coil.

The electrical and mechanical assemblies and parts of the coil must be used with care and should be routinely inspected.

Before using the coil, visually check it to ensure there is no external damage. Do not use the coil if the housing or cable is broken.

1-5-7 Modifications

Any changes or modifications to the coil must be approved and performed by GE Medical Systems prior to installation.

1-5-8 Reporting of Incidents

Users should contact GE Medical Systems immediately to report an incident and/or injury to a patient, operator or maintenance employee that occurred as a result of coil operation.

If an accident occurs as a result of coil operation, do not operate the equipment until an authorized investigation is conducted.

1-6 Shipping Contents

Box #	Part Name	GE Part #	USAI Part #	Qty
1	Coil	2380634-2	100410	1
2	Phantom Kit	2377426-3	550011	1
3	Coil Pad	2380634-5	150504	1
3	Ear Pad	2380634-6	150503	2
3	Ear Pad Covers	2380634-8	150537	1
3	Knee Pad	2380634-9	150529	1
1	Operator's Guide	2380634-3	770053	1
1	Service Manual	2380634-4	780049	1

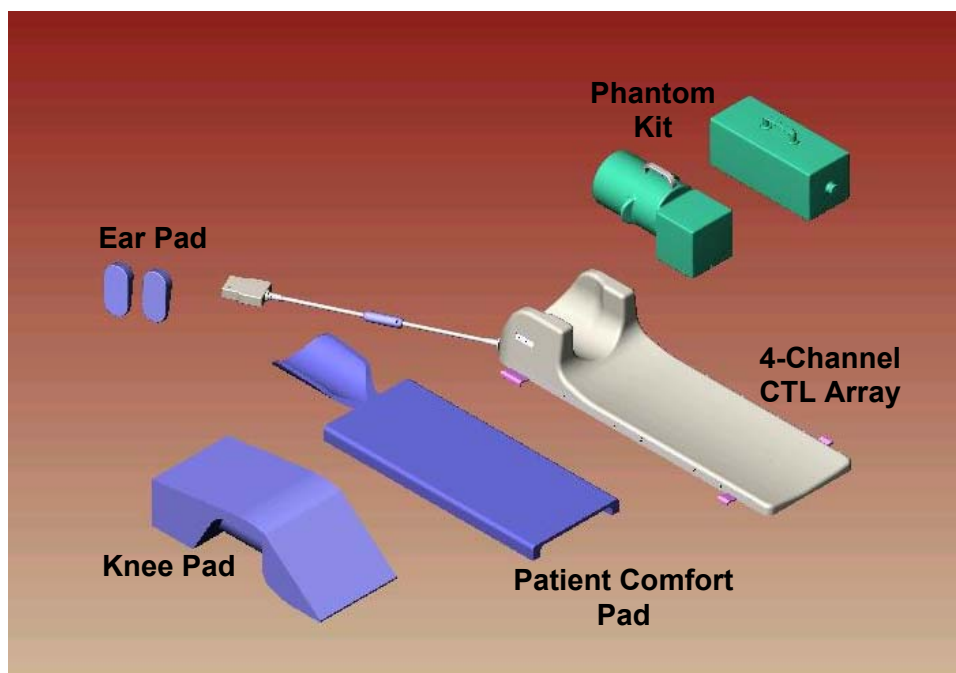


Figure 1: Parts of the coil.

1-7 Coil Description

The 4-Channel CTL Array is a receive-only coil designed to give optimum signal to noise ratio and uniform coverage of the spine anatomy, including the cervical, thoracic and lumbar spine regions.

The coil allows the user to obtain high-resolution images of extended regions (such as the thoracic-lumbar spine) or specific regions of interest (such as the cervical spine).

The coil has three modes of operation: (1) 4CTL12 for cervical imaging, (2) 4CTL23 for thoracic imaging, and (3) 4CTL34 for lumbar imaging.

These modes are enabled by the clinical user at the operator console. Other modes and configurations may be provided by GE Medical Systems.

1-8 Environmental Requirements

Store the coil in an air-conditioned scan room or equipment room.

ENVIRONMENTAL REQUIREMENTS – TABLE 1-8

	Transport and Storage	Operating Conditions
Barometric Pressure	645mm Hg to 795mm Hg	
Humidity	20% to 95% non-condensing	45% to 75% non-condensing
Temperature	-15° C to 55° C (5° F to 131° F)	15° C to 35° C (59° F to 95° F)

1-9 Coil Specifications

COIL SPECIFICATIONS – TABLE 1-9

Field Strength	Coil Model	GEMS Part #	Frequency
3.0T	4-Channel CTL Array	2380634-2	127.73 MHz

1-10 Coil Installation and Service

Coil installation and configuration requires the services of your GE Service Representative.

Please refer to the Service Manual (GE part number, 2380634-4) for coil installation and configuration information.

1-10-1 Coil Installation Procedure

Coil installation and configuration requires the services of your GE Service Representative.

Please refer to the Service Manual (GE part number, 2380634-4) for coil installation and configuration information.

1-10-2 Coil Service

Coil service requires the services of your GE Service Representative.

Please refer to the Service Manual (GE part number, 2380634-4) for information.

1-10-3 Replaceable Accessories**REPLACABLE ACCESSORIES – TABLE 1-10-3**

Part Name	GE Part #	USAI Part #
Coil Pad	2380634-5	155213
Ear Pad	2380634-6	155214
Ear Pad Covers	2380634-8	155216
Knee Pad	2380634-9	155215

For additional information, contact:

GE Medical Systems – Americas

USA 800-558-5102

Canada 800-668-0732

GE Medical Systems – Asia

India 91-80-845-2923

Japan 81-120-48-2630

China 86-21-621-92228

Australia 61-2-9975-5501

Singapore 65-291-8528

Taiwan 886-2-2505-7900

GE Medical Systems – Europe

33-1-41-19-7676

GE on the World Wide Web

www.gemedicalsystems.com/rad/mri/index.html

SECTION 2 – SETUP

2-1 Positioning the Coil

CAUTION

Remove any other coil or unused accessory device from the magnet before using the coil. Patient burns may result from coupling.

Step 1 – Position the coil on the cradle near the bore of the magnet lining up the table locators on the coil with the grooves in the table (see *Figure 2*).

Step 2 – Plug the coil connector into the RF coil receptacle. The coil connector and cable should always be routed directly to the magnet and should not form loops.



Figure 2: Positioning the coil on the cradle.

2-2 Positioning the Patient

Step 1 – Place a clean cotton sheet over the coil and comfort pad such that the patient's skin will not come in contact with the coil.

Step 2 – Position the patient head-first, supine, with the shoulders snugly resting against the coil sides on the coil (see *Figure 3*). For the feet-first scan, position the patient feet-first, supine, with legs resting on the knee pad on the coil (see *Figure 4*).



Figure 3: Positioning the patient on the coil.



Figure 4: Positioning the patient feet-first.

Step 3 – Provide earplugs to the patient after all instructions have been given.

WARNING!

Hearing protection is required for all people in the scan room during a scan to prevent hearing impairment. Acoustic levels may exceed 99 dB(A).

Hearing protection must have a Noise Reduction Rating (NRR) of 28dB or better (e.g., 30 dB, 32dB, etc.).

Step 4 – Put the ear pads between patient's ears and the coil former in order to keep the patient's head steady during the scan (see *Figure 5*).



Figure 5: Positioning the ear pads.

2-3 Localizing Anatomy

Position the alignment light on the anatomy of interest and landmark. The ruler on the coil indicates the physical lengths of Sections 1, 2, 3 and 4 (see *Figure 6*). Section 1 is the coverage for Mode 4CTL1. Sections 1 and 2 for Mode 4CTL12. Sections 2 and 3 for Mode 4CTL23. Sections 3 and 4 for Mode 4CTL34 and Section 4 for Mode 4CTL4. Laser mark the patient to the center of the coil coverage of the selected scan mode.

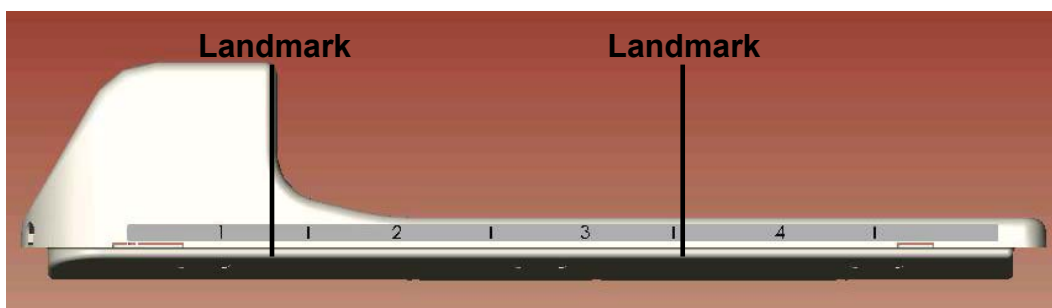


Figure 6: Landmarking for cervical and lumbar scanning (exact locations may change for some patients).

2-4 Recommended Scan Protocols

Use the System Recommended Protocols for scanning with this coil for optimal results and best performance.

Specific choice of protocols is entirely the responsibility of the radiologist and/or technologist.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1

Description	GE Part #	USAI Part #	Qty
Phantom Kit	2377426-3	550011	1

3-2-2 Explanation of Procedure

To fully test SNR on the 4-Channel CTL Array, two sections will be scanned. These sections are identified as 4CTL12 and 4CTL34. SNR measurements must be made for each section, requiring two sets of signal and noise scans in the coronal plane.

Refer to the Data Sheet in Section 3-2-6 to understand the data required to calculate the composite SNR for each section of the coil.

The image quality check uses two different protocols for signal and noise image acquisition. The signal scan is an **FSE** sequence used to minimize susceptibility and B₀ inhomogeneity effects. The noise scan is a **GRE** sequence that has a Control Variable (do_noise) to eliminate the transmit RF completely during the scan. The signal scan **must** be run prior to the noise scan as the R1, R2, and TG values from the signal scan are used for the noise scan.

3-2-3 Signal Scan

1. From the **Scan Desktop**, start new scan by selecting [**New Pt**]; set **Patient ID** to “geservice” and Patient Weight to “111” pounds. Click [**Patient Position**] to open protocols window.
2. Position the phantom set on the coil without the patient pad (see *Figure 7*).
3. Landmark on the vertical line corresponding to the region of interest (see *Figure 7*).



Figure 7: Landmarking for signal and noise scans.

4. At the magnet, press “**Alignment Light**” to turn on the light. Move the cradle to align the coil to the alignment lights. Press “**Landmark**” to landmark the alignment.
5. Move the coil to scan position by clicking on the “Move to Scan” button.
6. At the console, verify the coil has been properly identified by the system: correct picture on the screen and name in the coil field. If the system does not recognize the coil, refer to Section 2 - Setup and Calibration in the Service Manual.
7. Set the protocols per the Signal section from Table 3-2-4 Signal and Noise Protocols.
8. Select [**Save Series**] the [**Prepare to Scan**].
9. Prior to the scan, open the [**Display CVs**] menu under [**Research Operations**].
10. Set the “**saveinter**” CV to 1, saving the intermediate images. This is required to ensure all channels are operating well.
11. Select [**Auto Prescan**].
12. Record the R1, R2 and TG values obtained, then select [**Scan**].
13. If not continuing to noise scan, be sure to end the exam when finished. This will ensure that all CVs return to their default values.

3-2-4 Noise Scan

A signal scan must be run **prior** to the noise scan as the same R1, R2 and TG values must be used for both the signal and noise scans. Do **not** run an Auto Prescan prior to the noise scan as the values will be changed.

1. Copy the signal scan series. Use **[Copy Series]** (highlight signal series and click right mouse button) and **[Paste Series]** in **RX Manager**.
2. Click **[View Edit]** and set the protocols per the Noise section from Table 3-2-4 Signal and Noise Protocols.
3. Click **[Save Series]** and click **[Prepare to Scan]**.
4. Open the **[Display CVs]** menu under **[Research Operations]**.
5. To obtain the noise images, set the “**rhformat**” and “**do_noise**” CVs to 1.
6. Start **[Manual Prescan]**, look at the power spectrum and verify R1, R2, TG and Bandwidth have not changed from the previously completed signal scan.
7. Select **[Done]** and exit without changing any parameters.
8. Select **[Scan]**.

SIGNAL AND NOISE PROTOCOLS – TABLE 3-2-4

	Signal	Noise
Patient/Exam Information		
Patient ID	geservice	geservice
Patient Name	CTL Array	CTL Array
Patient Weight	111 lbs. (50 kg)	111 lbs. (50 kg)
Patient Position	Supine	Supine
Patient Entry	Head First	Head First
Coil	4CTL12/4CTL34	4CTL12/4CTL34
Series Description	Signal	Noise
Imaging Parameters		
Plane	Coronal	Coronal
Mode	2D	2D
Pulse Seq	FSE	GRE
Imaging Options	none	none
PSD Name	<i>leave blank</i>	<i>leave blank</i>
Protocol	<i>leave blank</i>	<i>leave blank</i>
Scan Timing		
# of Echoes	1	1
TE	17	minful
TR	500	34
Echo Train Length/Flip Angle	4	1
Bandwidth	15.63	15.63
Additional Parameters		
<i>no entries required in this area</i>		
Acquisition Timing		
Freq	256	256
Phase	256	256
NEX	1	1
Phase FOV	1.00	1.00
Preq DIR	S/I	S/I
Auto Center Freq	Peak	Peak
Autoshim	On	On
Phase Correct	On	On
Contrast	Off	Off
# of Reps B4 Pause	0	0
Scanning Range		
FOV	24	24
Slice Thickness	3	3
Spacing	1.5	1.5

	I/S Center	P/A Center	R/L Center
Start	0	0	
End	0		
# Slices	1	Table Delta	0.0

3-2-5 SNR Image Analysis

Image Analysis

Regions of interest in both signal and noise images can be measured directly in the image browser. Click the user interface button **Measure**, select the square or rectangular shape, and adjust its size and orientation when the shape is displayed in the selected image. Mean, standard deviation, and area of the ROI will appear in the lower right corner of the image.

Examples of typical intermediate Receiver Images for 4CTL12 are shown in *Figure 8*. Typical composite and noise images for 4CTL12 are shown in *Figures 9 and 10*.

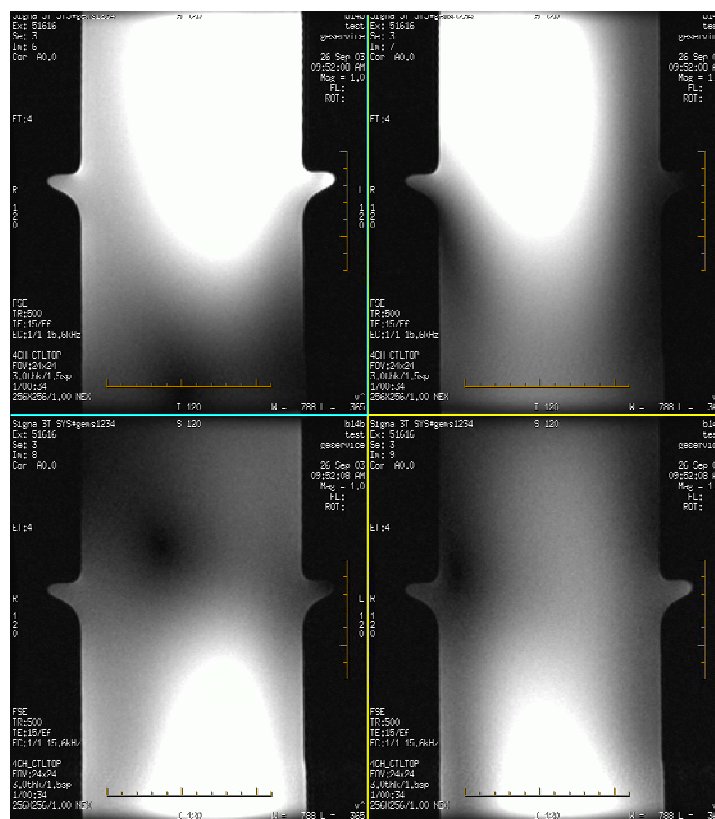


Figure 8: 4CTL12 intermediate receiver images 1-4.

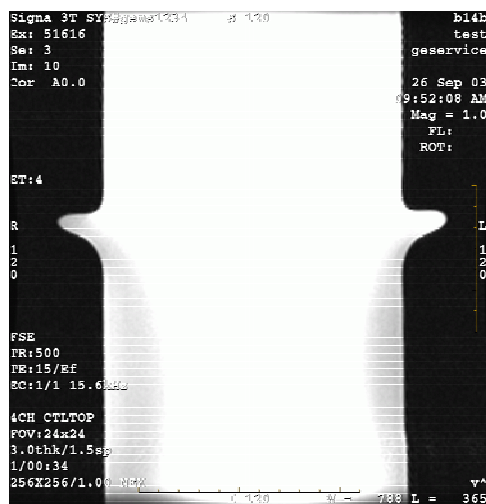


Figure 9: 4CTL12 composite image.



Figure 10: 4CTL12 noise image.

Examples of typical intermediate Receiver Images for 4CTL34 are shown in *Figure 11*. Typical composite and noise images for 4CTL34 are shown in *Figures 12 and 13*.

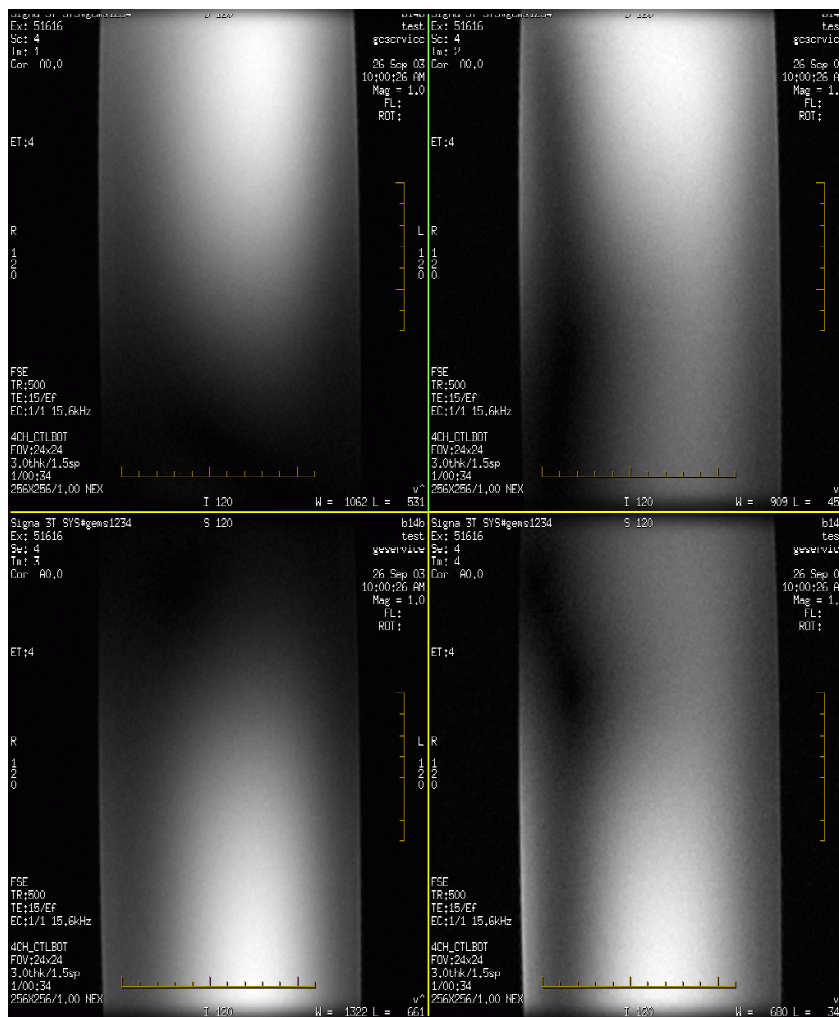


Figure 11: 4CTL34 intermediate receiver images 1-4.

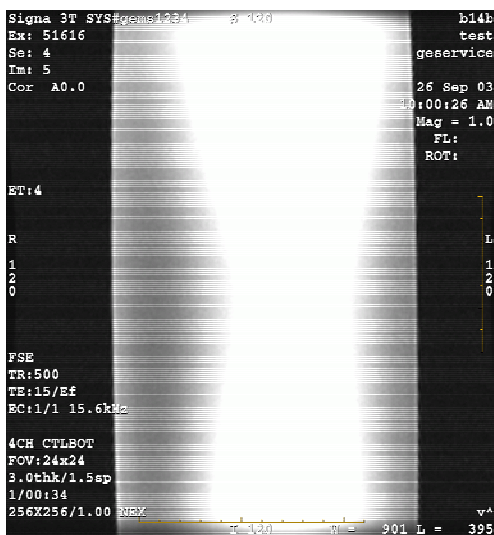


Figure 12: 4CTL34 composite image.

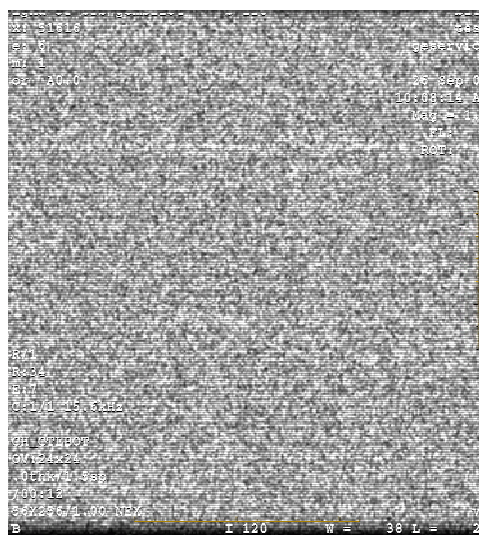


Figure 13: 4CTL34 noise image.

SNR Measurement

To acquire the composite signal image, take two SNR measurements for 4CTL12. A 10cm square ROI should be used for the top two set of elements in the cervical section of the coil (see *Figure 14*). For the Noise image, a 24cm square ROI should be used (see *Figure 15*).

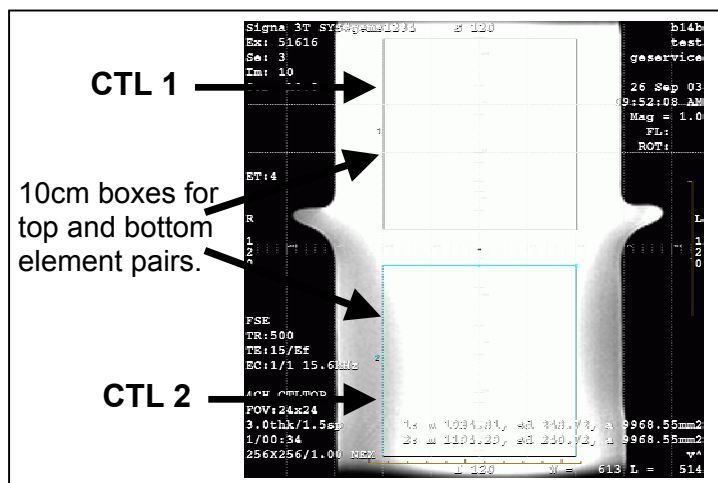


Figure 14: Composite 4CTL12 SNR ROIs.

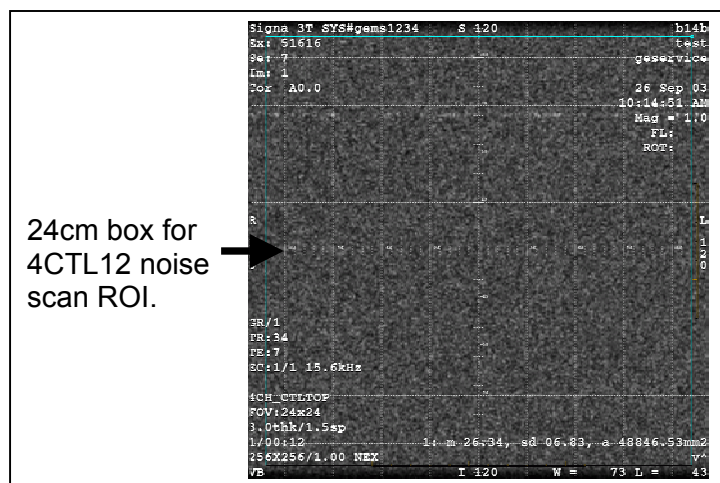


Figure 15: 4CTL12 noise ROI.

To acquire the composite signal image, take one SNR measurement for 4CTL34. A rectangular 20cm height and 10cm width ROI should be used for the bottom two sets of elements in the lumbar section of the coil (see *Figure 16*).

For the noise image, a 24cm square ROI should be used (see *Figure 17*).

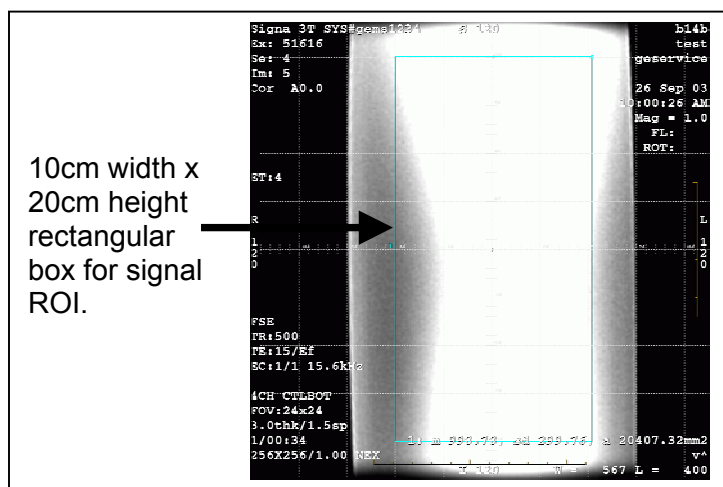


Figure 16: Composite 4CTL34 SNR ROI.

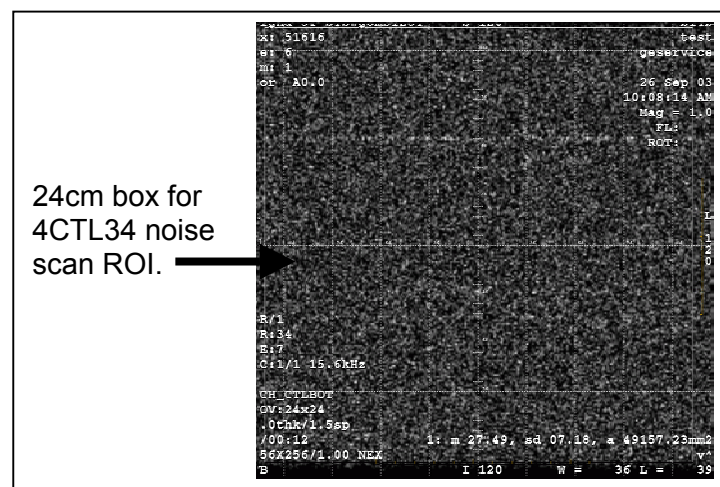


Figure 17: 4CTL34 noise ROI.

The SNR is calculated using the signal to noise ratios of the individual receiver channels:

$$\text{SNR}_{\text{Composite}} = \frac{\text{Mean of Signal within ROI}}{\text{Standard Deviation of Noise within ROI}}$$

Note: The SNR calculation uses the **MEAN** of the signal image and **STANDARD DEVIATION** of the noise image. SNR is measured on the composite image.

SNR Specification

The SNR measurements must be greater than or equal to the following specifications:

SNR SPECIFICATIONS – TABLE 3-2-5

Coil Element	Composite SNR
CTL1	240
CTL2	145
CTL34	117

3-2-6 SNR Data Sheet

Use the table provided below to record the calculated signal to noise ratio (SNR) data obtained from the Functional Checks section.

Date	Comments						
Mode	R1	R2	TG	Composite Image Signal Mean	Noise Std Dev	SNR	Spec Limit
4CTL1							
4CTL2							
4CTL34							

Date	Comments						
Mode	R1	R2	TG	Composite Image Signal Mean	Noise Std Dev	SNR	Spec Limit
4CTL1							
4CTL2							
4CTL34							

SECTION 4 – MAINTENANCE

4-1 Coil Care

WARNING!

Detach coil connector from the scanner before attempting to clean. Do not reattach after cleaning until the coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.

CAUTION

4-1-1 Coil Care

The 4-Channel CTL Array and patient comfort pads must be cleaned and stored using the following procedure. Wipe with a cloth that has been dampened in a solution of 10 percent bleach and 90 percent tap water, or 30 percent isopropyl alcohol and 70 percent tap water.

Do not spray or pour any cleaning solution directly on the coil!

Store the coil in an air-conditioned scan room or equipment room.

Under no circumstances should the coil be placed into any type of sterilizer.

4-1-2 Storage Requirements

Store the coil in an air-conditioned scan room or equipment room.

To store the coil, a storage space exceeding the coil dimensions of 43.3" L x 16.1" W x 9.3" H is required.

REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
1	11/03	Teresa DeMarco	First Issue